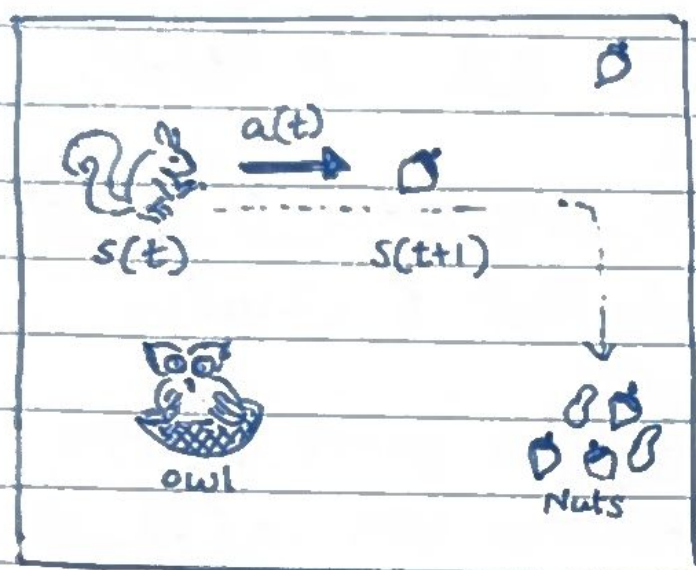


# Reinforcement Learning

Consider a squirrel foraging for nuts. Some parts of the environment will have more nuts, and some other parts may have threats, for instance an owl's nest. By exploring and interacting with the environment, the squirrel learns to move through space to reach regions with nuts while avoiding threats.

Learning what to do by interacting with the environment is the focus of Reinforcement Learning



Agent: Learner & decision maker.

Environment: What the agent interacts with.

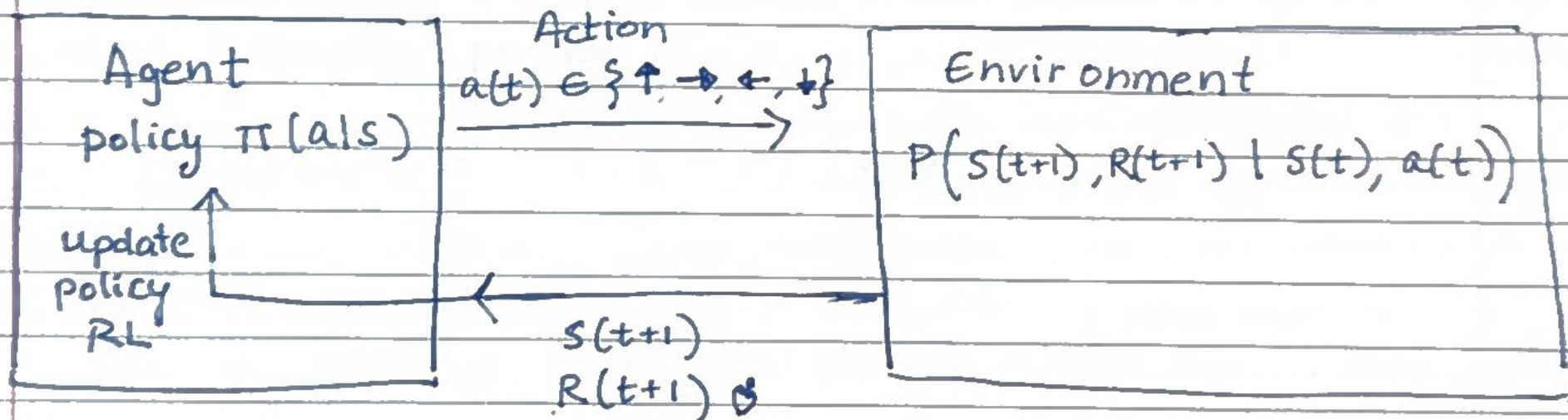
$S(t)$ : Representation of environment's state at time  $t$ .  $s \in S$

$a(t)$ : Action the agent takes at time  $t$ .

$R(t)$ : Reward the agent receives at time  $t$ .

Markov property

$R(t+1)$  &  $S(t+1)$  depend only on  $s(t)$  and  $a(t)$ , not on earlier states  $s(t-1)$ ,  $s(t-2)$  ... and earlier actions  $a(t-1)$ ,  $a(t-2)$  ...





Goal: To maximize Expected discounted Return

$$Q(t) = R(t+1) + \gamma R(t+2) + \gamma^2 R(t+3) + \dots$$

$$0 < \gamma < 1$$

if  $\gamma = 0$   $Q(t) = R(t+1)$  Myopic agent

if  $\gamma = 1$   $Q(t) = R(t+1) + R(t+2) + R(t+3) + \dots$

Far sighted agent.

~~~~~  
Value function of State  $s$  under policy  $\pi$

$$V_{\pi}(s) = E_{\pi}[Q(t) | s(t)=s]$$

Policy  $\pi$  better than or equal to policy  $\pi'$  if and only if

$$V_{\pi}(s) \geq V_{\pi'}(s) \quad \forall s \in \mathcal{S}$$

Optimal policy  $\pi^*$

$$V_{\pi^*}(s) = \max_{\pi} V_{\pi}(s) = E_{\pi^*}[Q(t) | s(t)=s]$$

~~~~~  
State - Action Value function

Expected return for starting at state  $s$  and taking action  $a$ , and then following policy  $\pi$

$$Q_{\pi}(s, a) = E_{\pi}[Q(t) | s(t)=s, a(t)=a]$$

Optimal policy  $\pi^*$

$$Q_{\pi^*}(s, a) = E_{\pi^*}[Q(t) | s(t)=s, a(t)=a] = \max_{\pi} Q_{\pi}(s, a)$$



## Bellman Optimality Equation for $Q_{\pi^*}(s, a)$

$$Q_{\pi^*}(s, a) = E_{\pi^*}(Q(t) | s(t)=s, a(t)=a) \Rightarrow \textcircled{1}$$

$$Q(t) = R(t+1) + \gamma R(t+2) + \gamma^2 R(t+3) + \dots$$

$$Q(t+1) = R(t+2) + \gamma R(t+3) + \dots$$

$$Q(t) = R(t+1) + \gamma Q(t+1) \Rightarrow \textcircled{2}$$

Substituting  $\textcircled{2}$  in  $\textcircled{1}$ .

$$Q_{\pi^*}(s, a) = E_{\pi^*}(R(t+1) + \gamma Q(t+1) | s(t)=s, a(t)=a)$$

$$Q_{\pi^*}(s, a) = \sum_{s', r} P(s', r | s, a) (r + \gamma E[Q(t+1) | s(t+1)=s'])$$

$$Q_{\pi^*}(s, a) = \sum_{s', r} P(s', r | s, a) (r + \gamma V_{\pi^*}(s')) \Rightarrow \textcircled{3}$$

$$V_{\pi^*}(s') = \max_{a'} Q_{\pi^*}(s', a') \Rightarrow \textcircled{4}$$

Under an optimal policy  $\pi^*$ , Expected return of state  $s'$  should be the expected return for the best action from that state.

Substituting  $\textcircled{4}$  in  $\textcircled{3}$

$$Q_{\pi^*}(s, a) = \sum_{s', r} P(s', r | s, a) (r + \gamma \max_{a'} Q_{\pi^*}(s', a'))$$

$$Q_{\pi^*}(s, a) = E[R(t+1) + \gamma \max_{a'} Q_{\pi^*}(s(t+1), a')]$$

## Bellman Optimality Equation for $Q_{\pi^*}$

Satisfies recursive relationship

Given the dynamics of the environment  $P(s', r | s, a)$ , one can solve for  $Q_{\pi^*}(s, a)$

Then we can find the next action that maximizes expected discounted return by

$$a^* = \arg \max_a Q_{\pi^*}(s, a)$$



## Q-Learning

Learned action value function  $Q(s, a)$

We want it to approximate  $Q_{\pi^*}(s, a)$

$$Q_{\pi^*}(s, a) = E \left[ R(t+1) + \gamma \max_{a'} Q_{\pi^*}(s(t+1), a') \right]$$

Bellman Error

$$R(t+1) + \gamma \max_{a'} Q(s(t+1), a') - Q(s, a)$$

Update  $Q(s, a)$  to minimize Bellman Error

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left( R(t+1) + \gamma \max_{a'} Q(s(t+1), a') - Q(s, a) \right)$$

$\alpha$ : learning rate

$Q$  approximates  $Q_{\pi^*}$  independent of policy being followed.

Agent only updates  $Q$  for states and actions it visits

$\Rightarrow$  Needs to explore new states, actions

Exploration

—

Exploitation

Tradeoff

$\Downarrow$

$\epsilon$  times

$\Downarrow$

$1 - \epsilon$  times

Pick suboptimal actions  
to visit new states and  
new actions

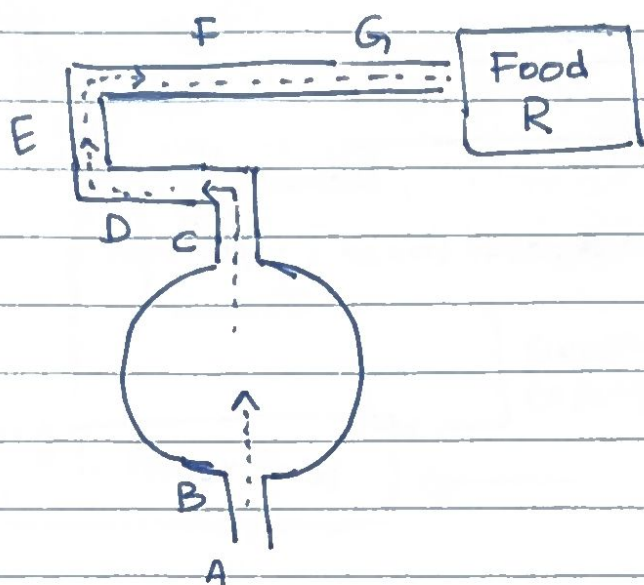
Choose action  
according to  $Q$   
 $a^* = \arg \max_a Q(s, a)$

$\epsilon$ -greedy policy

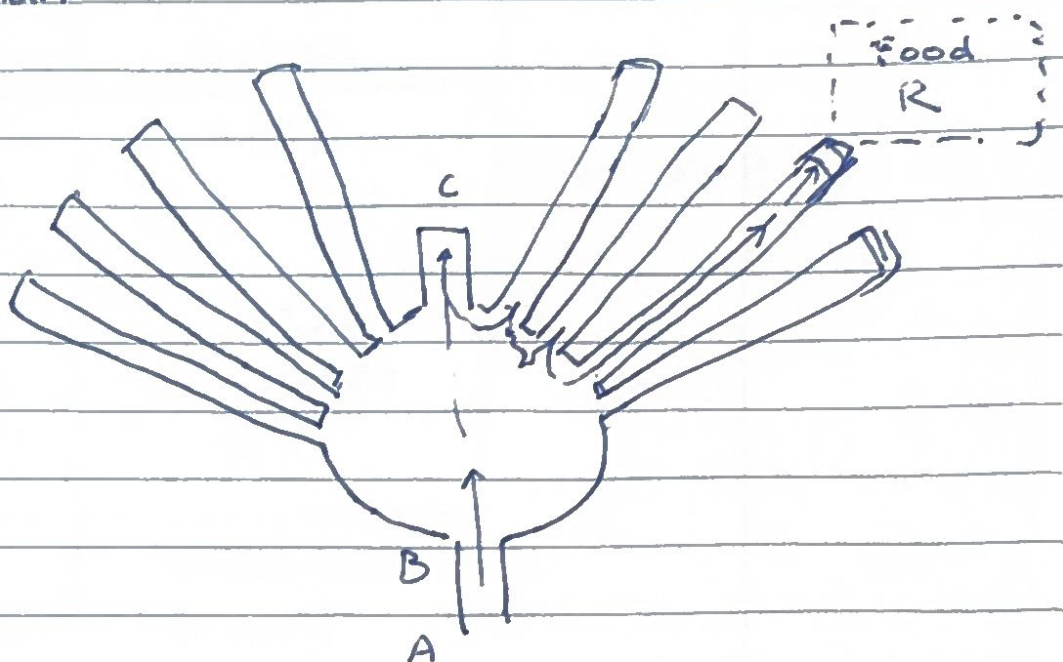


# Evidence for Generative Model

## Experiments by Edward Tolman



Original Maze



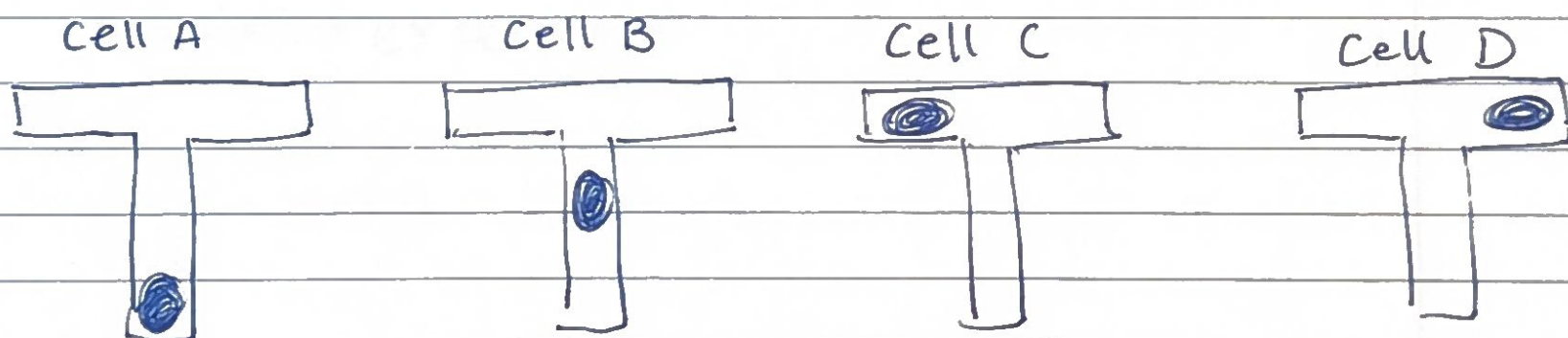
Modified maze

With just learning  $Q(s,a)$ , an agent wouldn't know how to move in the modified maze, which suggests that animals may be learning a generative map or model of the environment.

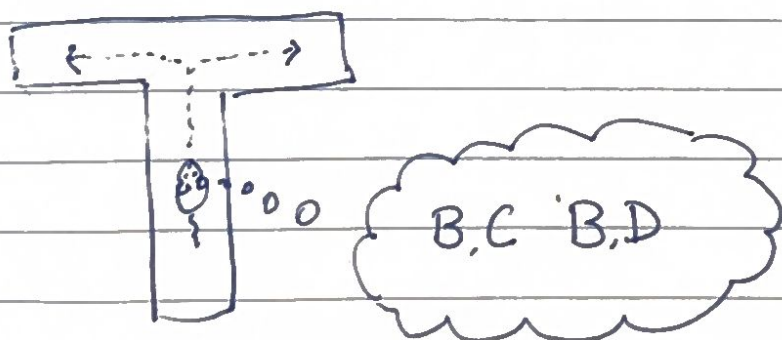
## Evidence for generative model and simulated Experience

### Hippocampus

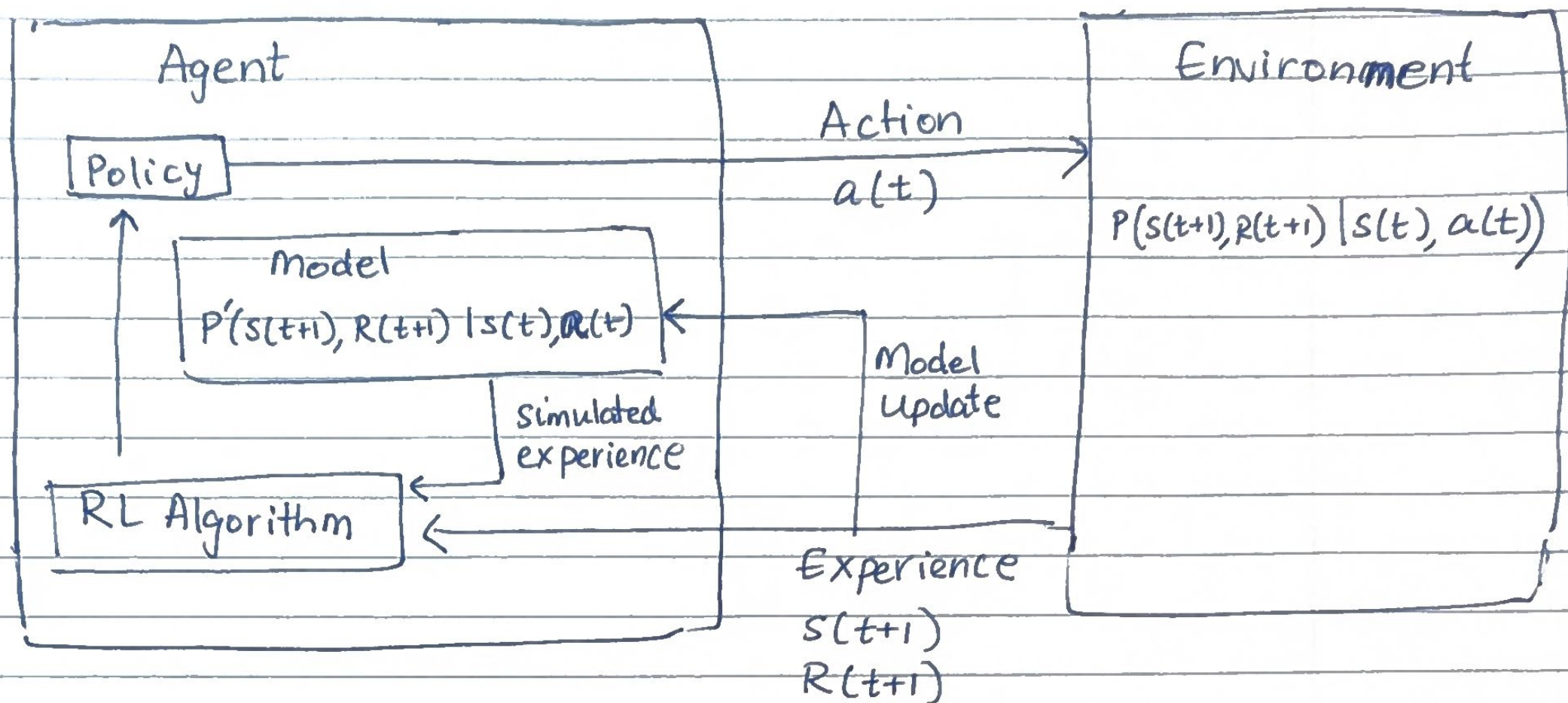
population activity  $\Rightarrow$  state of the environment



Planning

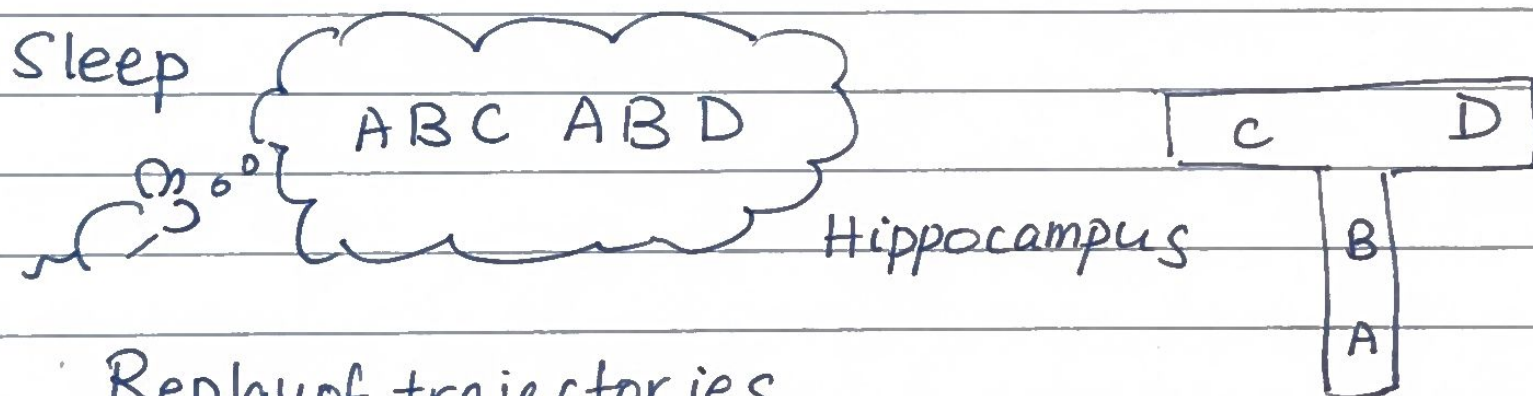


## Model-based RL



RL Agent can learn from simulated experience

Leads to better sampling efficiency



Replay of trajectories  
Simulated experience